

Capstone Project 1 Report

“Global Startup & Investment Intelligence Platform”

A Power BI-Driven Platform for Analyzing Startup Ecosystems, Funding Trends, Venture Capital Investments, Acquisitions, and IPO Performance

Domain: Startup & Investment Analytics

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1. Introduction

In today's global startup economy, investors, founders, and analysts face a growing challenge in understanding which sectors, regions, and funding strategies actually drive successful outcomes. This project focuses on analysing startup funding, investment, and exit data using data-driven techniques and building an interactive dashboard to assist decision-makers in identifying patterns in funding behaviour, sector performance, and global investment trends.

2. Problem Statement

The global startup ecosystem has expanded rapidly over the last decade, attracting hundreds of billions of dollars in investment across hundreds of thousands of companies. However, the data describing this activity — funding rounds, investor participation, acquisitions, and IPOs — is scattered across disconnected records. Identifying which sectors and regions are genuinely attracting capital, understanding investment patterns, and tracking KPIs like total funding and round volume are essential for evaluating opportunities. The lack of a visual, interactive, and centralized system makes it difficult for investors and analysts to make informed decisions.

3. Objective

The primary goal of this project is to:

- Analyse historical startup funding and investment data to extract meaningful insights.
- Visualize KPIs to monitor funding performance, round activity, and exit trends.
- Differentiate between high-growth and low-growth sectors and regions using filters and breakdowns.
- Help investors, founders, and analysts identify global investment trends and improve decision-making.

4. Tools & Technologies

Tool	Purpose
Python (Pandas)	Data cleaning, aggregation, and statistical validation of CSV records
Power BI	Visual dashboard creation and interactive reporting
Power Query & DAX	Data transformation, relationships, and KPI measure creation
Microsoft Excel	Quick data inspection and ad-hoc validation

CSV Files	Source data used for analysis (Crunchbase startup investment dataset)
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5. Methodology

1. Data Import & Cleaning:

- Imported five linked Crunchbase CSV datasets (objects, funding_rounds, investments, acquisitions, ipos) into Power BI.
- Filtered the objects table to isolate startup company records from people, products, and investor entities.
- Standardized country codes and category labels, and handled missing or blank funding values.

2. Data Transformation:

- Built relationships across the five tables using company and funding-round identifiers.
- Created funding categories, status groupings, and KPI measures using Power Query and DAX.

3. Validation in Python:

- Used Python (Pandas) to independently aggregate totals — total funding, startup counts, category and country breakdowns.
- Cross-verified these computed figures against the Power BI dashboard measures to confirm accuracy.

4. Dashboard Development (Power BI):

- Imported the cleaned and validated data into Power BI.
- Built visualizations including bar charts, donut charts, trend lines, geographic maps, and KPI cards.
- Added filters for year, country, category, and status.

6. Data Description

Source: Crunchbase Startup Investments Dataset (objects.csv, funding_rounds.csv, investments.csv, acquisitions.csv, ipos.csv)

Total Records: 196,553 startup companies (462,651 total entities including people, products, and investors), 52,928 funding rounds, 80,903 investment records, 9,562 acquisitions, and 1,259 IPOs.

Key Fields:

- **category_code, status, country_code** – Startup profile and classification attributes.
- **funding_total_usd, funding_rounds** – Aggregate funding metrics per startup.

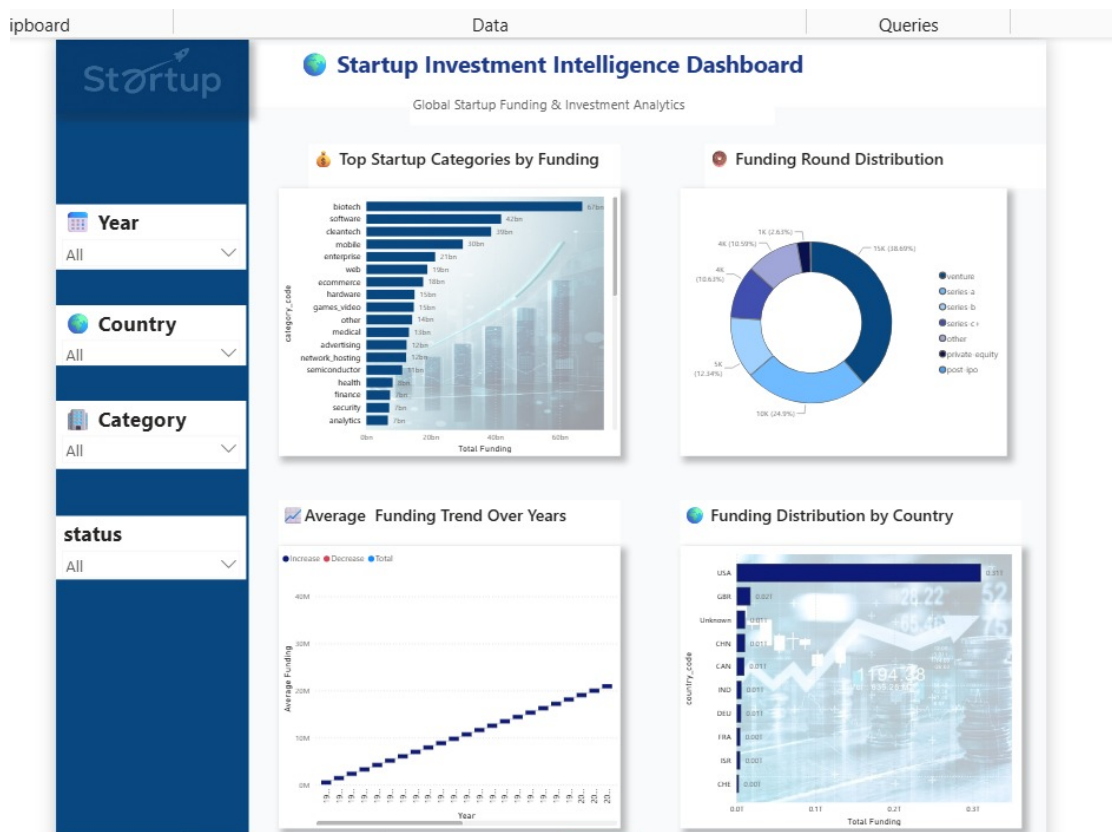
- **funding_round_type, raised_amount_usd, funded_at** – Individual funding round details.
- **investor_object_id, funded_object_id** – Investor – startup relationship records.
- **price_amount, acquired_at** – Acquisition transaction details.
- **valuation_amount, raised_amount, public_at** – IPO and public listing details.

7. Dashboard Design

Sections:

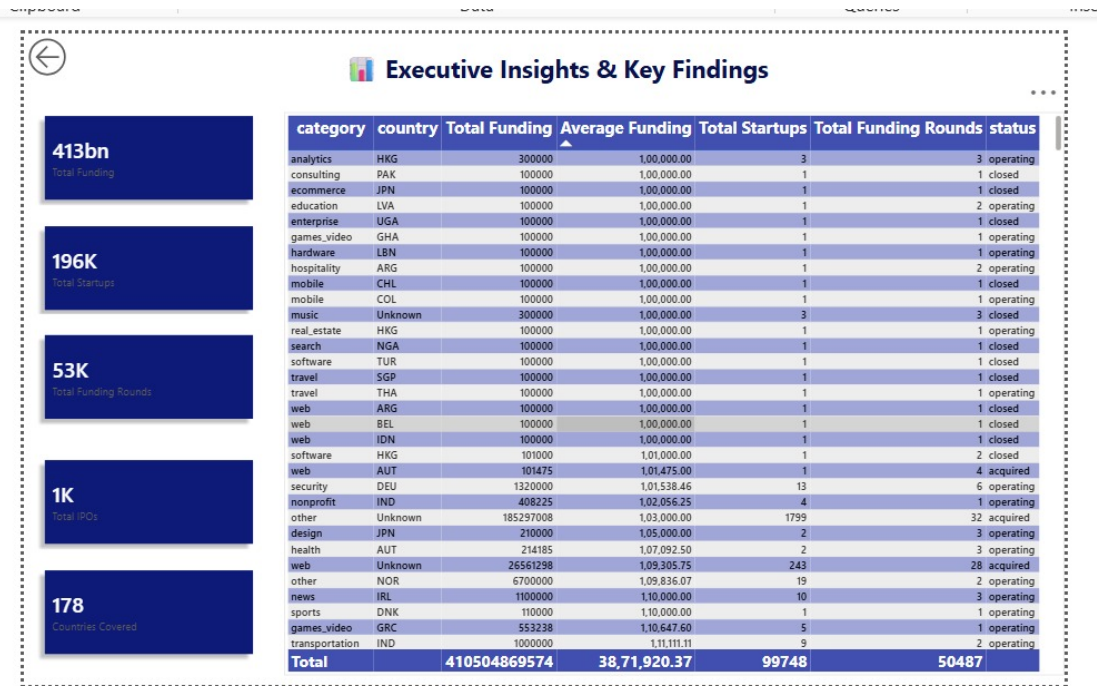
1. **KPI Cards:** Show total funding, total startups, total funding rounds, total IPOs, and countries covered.
2. **Category & Round Breakdown:** Top startup categories by funding and funding round type distribution.
3. **Trend Chart:** Average funding trend over the years.
4. **Geo Map & Bar Charts:** Country-wise funding distribution.
5. **Drill-Through Insights Table:** Category x country breakdown with total funding, average funding, startup count, round count, and status.
6. **Filters:** Enable slicing by year, country, category, and status.

Dashboard 1: Startup Investment Intelligence Dashboard



Overview dashboard showing top categories by funding, funding round distribution, average funding trend over the years, and funding distribution by country.

Dashboard 2: Executive Insights & Key Findings



Executive summary view with headline KPI cards (total funding, total startups, total funding rounds, total IPOs, countries covered) and a drill-through table breaking results down by category, country, and status.

8. Key Insights

1. **Biotech** is the leading category by total funding (\$66.9B), followed by software (\$41.8B) and cleantech (\$38.7B).
2. The **USA** dominates global funding volume (\$310.6B), far ahead of the UK (\$17.7B) and China (\$10.7B).
3. **Venture and angel rounds** together account for more than half of all recorded funding events.
4. The vast majority of startups (**over 93%**) remain in operating status, while only a small share are acquired, closed, or have gone public.
5. Average funding per round has trended **upward** across the years analysed.

9. Challenges Faced

- Inconsistent or missing country and category codes required standardization before aggregation.
- The objects.csv file contained multiple entity types (companies, people, products, investors) that had to be filtered to isolate startup company records.
- Reconciling KPI totals computed independently in Python against the Power BI dashboard required careful validation.

- Large file sizes (objects.csv alone exceeded 280MB) required efficient data handling to avoid performance issues.
- Imbalanced status distribution (operating startups vastly outnumbering acquired, closed, or IPO records) required careful chart scaling to avoid misleading visuals.

10. Real-Life Applications

This dashboard can be used by:

- **Venture Capital Firms:** To identify high-growth sectors and regions for new investment opportunities.
- **Startup Founders:** To benchmark funding expectations against peers in the same category and stage.
- **Investment Analysts:** To track funding trends and exit performance across global markets.
- **Policy Makers:** To understand which regions and industries are driving innovation and economic activity.
- **Data Analysts:** As a base model to expand into predictive funding or exit-prediction models.

11. Conclusion

This project demonstrates how raw, fragmented startup investment data can be transformed into a meaningful, visual story that supports investment and strategic decision-making. Using Python and Power BI, the Global Startup & Investment Intelligence Platform provides a robust tool for tracking funding performance, analysing sector and regional trends, and monitoring exit activity. It acts as a bridge between raw data and strategic insight for investors, founders, and analysts alike.

12. Future Scope

1. **Predictive Modelling:** Use machine learning to predict which startups are likely to raise follow-on funding or exit successfully.
2. **Real-Time Data Integration:** Connect the dashboard to live funding databases for real-time monitoring.
3. **Investor Network Analysis:** Map investor co-investment patterns and syndicate behaviour.
4. **Advanced Metrics:** Add financial KPIs like internal rate of return (IRR) and valuation multiples.
5. **Role-Based Dashboards:** Customize views for venture capitalists, founders, and analysts.
6. **Sector Deep-Dives:** Extend category-level analysis into sub-industry breakdowns.
7. **Multi-Year Forecasting:** Analyse trends across multiple years to forecast future funding cycles.
8. **Mobile Optimization:** Deploy the dashboard on Power BI mobile for anytime, anywhere access.

13. References

<https://www.kaggle.com/datasets/justinas/startup-investments>